

BUHTA AMBARCHIK, Russian Federation

WMO: 250340

Lat: 69.624N

Lon: 162.298E

Elev: 23

StdP: 101.05

Time Zone: 11.00 (E11)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-40.8	-39.5	-42.7	0.1	-40.5	-41.2	0.1	-39.2	17.7	-27.8	16.5	-25.5	3.9	230	1.057

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.7	21.6	15.0	18.9	13.4	16.3	12.2	15.8	20.4	14.0	18.4	12.2	16.0	5.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.7	9.8	17.7	11.6	8.5	16.5	9.8	7.5	14.4	44.5	20.8	39.3	18.5	34.9	16.1	20.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.5	14.6	12.9	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-10.3	-28.4	-28.6	-23.2	-12.5	-3.4	4.8	7.8	6.3	2.4	-7.3	-17.6	-25.2		
	DBStd	14.75	6.76	7.39	7.04	5.90	5.42	5.47	4.71	4.96	3.84	5.95	7.70	7.81		
	HDD10.0	7502	1191	1080	1029	677	416	180	103	141	231	536	827	1091		
	HDD18.3	10457	1449	1313	1287	927	674	407	327	374	477	794	1077	1349		
	CDD10.0	89	0	0	0	0	0	23	36	26	4	0	0	0		
	CDD18.3	3	0	0	0	0	0	1	2	1	0	0	0	0		
	CDH23.3	23	0	0	0	0	0	2	17	3	0	0	0	0		
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0		
	WSAvg	5.5	6.1	5.9	5.4	4.6	4.9	5.2	4.9	5.5	5.1	5.3	6.2	6.9		
	PrecAvg	137	5	4	5	4	6	14	24	22	21	16	10	7		
(15) Precipitation	PrecMax	227	14	11	16	9	18	36	56	50	59	38	20	30		
	PrecMin	50	0	0	0	0	1	1	1	8	2	3	3	1		
	PrecStd	51	4	3	4	3	5	11	17	12	14	8	5	6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.0	-3.0	-3.2	2.5	13.0	23.4	25.8	23.0	18.9	5.4	0.7	1.4		
		MCWB	-3.0	-5.1	-4.0	0.7	7.0	13.4	16.8	18.0	11.6	3.8	-0.8	0.0		
	2%	DB	-9.8	-9.4	-7.9	0.6	8.7	20.6	22.2	19.7	13.6	3.1	-1.3	-7.3		
		MCWB	-10.4	-9.9	-8.7	-1.0	5.2	12.6	15.6	15.5	9.4	2.3	-2.0	-8.0		
	5%	DB	-16.6	-16.2	-10.1	-1.8	5.7	17.8	19.2	17.0	10.0	1.8	-3.2	-11.7		
		MCWB	-17.1	-16.6	-10.9	-3.1	3.2	11.4	14.0	13.3	7.4	1.4	-4.1	-12.0		
	10%	DB	-19.6	-19.7	-12.5	-3.9	3.7	14.7	16.3	14.1	7.6	0.3	-6.4	-14.6		
		MCWB	-19.9	-20.0	-13.0	-5.0	1.7	9.9	13.1	11.3	5.6	-0.1	-7.2	-14.9		
	0.4%	WB	-3.3	-4.8	-3.9	1.1	7.6	14.4	17.4	18.9	11.9	4.1	-0.7	-0.4		
		MCDB	-1.6	-3.4	-3.1	2.3	12.3	21.8	23.3	22.3	17.6	4.8	0.3	0.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-10.5	-10.2	-8.6	-0.9	5.3	12.9	15.9	16.4	9.6	2.4	-1.9	-7.9		
		MCDB	-10.0	-9.4	-7.9	0.8	8.8	19.6	21.3	19.1	12.8	2.9	-1.1	-7.1		
	5%	WB	-17.0	-16.5	-10.9	-3.0	3.2	11.5	14.8	13.2	7.7	1.4	-4.1	-12.0		
		MCDB	-16.7	-16.1	-10.1	-1.8	5.5	17.6	18.8	16.5	9.8	1.8	-3.2	-11.8		
	10%	WB	-19.9	-19.9	-13.0	-4.9	1.8	9.9	13.1	11.3	5.7	-0.1	-7.2	-14.9		
		MCDB	-19.7	-19.7	-12.4	-3.9	3.3	14.7	16.4	13.9	7.4	0.3	-6.4	-14.6		
	5% DB	MDBR	5.7	4.5	5.6	6.8	5.5	6.1	6.7	5.7	4.3	4.0	5.8	5.3		
		MCDBR	11.1	9.5	7.9	8.4	7.8	12.0	13.1	10.6	8.1	3.8	6.8	9.0		
		MCWBR	10.4	9.2	7.5	7.3	5.1	6.7	7.9	6.9	5.4	3.5	5.8	8.2		
		MCWBR	11.0	9.4	8.1	7.9	7.5	11.6	12.4	9.8	7.4	3.8	6.5	8.5		
(35) Clear-Sky Solar Irradiance	taub		0.164	0.212	0.285	0.330	0.358	0.350	0.397	0.382	0.313	0.247	N/A	N/A		
		taud	1.907	2.028	2.042	1.973	2.053	2.288	2.269	2.304	2.446	2.117	N/A	N/A		
	Edn at Noon		33	538	704	781	810	839	776	739	710	466	0	0		
		Edh at Noon	3	41	85	126	132	108	108	93	60	37	0	0		
(44) All-Sky Solar Radiation	RadAvg		0.01	0.37	2.06	4.49	6.22	6.15	4.94	3.34	1.68	0.53	0.00	0.00		
	RadStd		0.00	0.06	0.10	0.15	0.36	0.43	0.41	0.31	0.13	0.07	0.00	0.00		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)		+1.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page